IGCSE 0607 Practice Questions — E2.7

Sequences

Total Questions: 18 — **Total Marks:** 111 Marks

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1 Here are the first four terms of a sequence.

5 12 19 26

(a) Write down the next term in the sequence.

..... **[1]**

(b) Find an expression for the nth term of the sequence.

..... **[2]**

(c) Explain why 150 is not a term in this sequence.

..... **[1]**

2 Here are the first four terms of a sequence.

5 11 17 23

(a) Find the next term in this sequence.

..... **[1]**

(b) Find an expression for the nth term of this sequence.

..... **[2]**

(c) Explain why 150 is not a term in this sequence.

..... **[1]**

3 The first four terms of a sequence are

7 11 15 19

(a) Write down the next two terms of the sequence.

..... **[1]**

(b) Find an expression for the nth term of this sequence.

nth term = ... **[2]**

(c) Explain why 102 is not a term in this sequence.

..... **[2]**

4 **(a)** Sequence A is a linear sequence.

5 9 13 17 ...

(i) Find the next two terms of sequence A.

..... **[1]**

(ii) Find the nth term of sequence A.

..... **[2]**

(b) Sequence B is a geometric sequence.

2 6 18 54 ...

Find the nth term of sequence B.

..... **[2]**

5 The first five terms of a sequence, T_n, are given below.

4 14 34 68 120

(a) By finding the first, second and third differences, show that the nth term of the sequence is a cubic expression in n.

[2]

(b) Find an expression for T_n, the nth term of the sequence.

$T_n =$ ..... **[5]**

6 **(a)** The nth term of a sequence is $n^2 - 3n + 5$.
Work out the first three terms of this sequence.

1st term = ... , 2nd term = ... , 3rd term = ...
[2]

(b) The first four terms of another sequence are

2 9 20 35

Find an expression for the nth term of this sequence.

..... **[3]**

7 The first four terms of a sequence, T, are

2 9 22 41

(a) Find an expression for the nth term, T_n.

$T_n =$ ..... **[3]**

(b) A second sequence is given by the formula $U_n = 2^n + 3$.
Work out the value of U_5.

$U_5 =$ ..... **[1]**

(c) Find the value of n for which $U_n = 131$.

$n =$ ..... **[2]**

8 The first four terms of a sequence are

$$-1 \qquad 2 \qquad 9 \qquad 20$$

(a) Find the nth term of this sequence.

..... **[3]**

(b) The nth term of another sequence is given by $3^n - 2$.
Find the value of the term in this sequence that is closest to 1000.

..... **[3]**

9 The first five terms of a sequence are

2 9 20 35 54 ...

(a) Find the next term.

..... **[1]**

(b) Find the nth term of the sequence.

..... **[3]**

(c) The kth term of the sequence is 464.
Find the value of k.

$k =$ ..... **[3]**

10 Consider the following three sequences, P, Q and R.
Sequence P:

$$2 \quad 9 \quad 28 \quad 65 \quad 126 \quad \ldots$$

Sequence Q:

$$3 \quad 8 \quad 15 \quad 24 \quad 35 \quad \ldots$$

Sequence R:

$$\frac{2}{3} \quad \frac{9}{8} \quad \frac{28}{15} \quad \frac{65}{24} \quad \frac{126}{35} \quad \ldots$$

(a) Find the nth term of sequence P.

..... **[2]**

(b) Find the nth term of sequence Q.

..... **[2]**

(c) Write down the nth term of sequence R.

..... **[1]**

(d) Find the 12th term of sequence R, giving your answer as a fraction in its simplest form.

..... **[2]**

11 **(a)** The first five terms of a sequence are

$$-1 \quad 8 \quad 33 \quad 80 \quad 155$$

Use the method of differences to find an expression for the nth term of this sequence.

..... **[4]**

(b) An exponential sequence has nth term $P_n = 5 \times 3^{n-1}$.
The kth term of this sequence is 3645.
Work out the value of k.

$k =$ ..... **[2]**

(c) The nth term of sequence T is $T_n = \dfrac{2n-1}{3n+2}$.
Write down the value that T_n approaches as n becomes very large.

..... **[1]**

12 **(a)** A sequence has the following first four terms:

1 15 53 127

Use the method of differences to find an expression for the nth term.

nth term = ... **[4]**

(b) Another sequence, V, has its nth term defined by

$$V_n = \frac{n^2 + 1}{2^n}$$

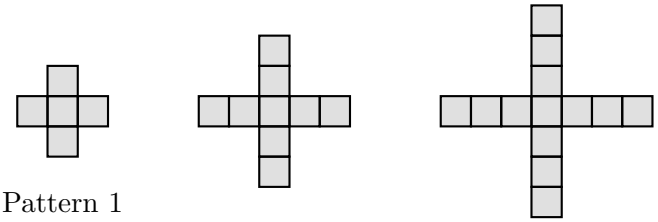
(i) Find the values of V_1, V_2 and V_3.
Give each fraction in its simplest form.

V_1 = ... , V_2 = ... , V_3 = ... **[2]**

(ii) Work out the exact value of $V_5 - V_4$.

..... **[3]**

13 The diagram shows a sequence of patterns made from small squares.



Pattern 1

Pattern 2

Pattern 3

(a) Find the number of squares in Pattern 4.

..... **[1]**

(b) Find an expression, in terms of n, for the number of squares in Pattern n.

..... **[2]**

(c) Which pattern has exactly 141 squares?

Pattern ..... **[2]**

14 The first four terms of a sequence are

$$-1 \qquad 4 \qquad 13 \qquad 26$$

(a) Find an expression for the nth term of the sequence.

..... **[3]**

(b) Find the value of n when the nth term is 433.

$n =$ ..... **[3]**

15 **(a)** The first five terms of a sequence are

4 5 14 37 80

Use the method of differences to find an expression for the nth term.

..... **[4]**

(b) The nth term of a different sequence is given by $T_n = a \times 2^n + b$.
The first term of this sequence is 10 and the third term is 34.

Find the values of a and b.

$a =$ ..... , $b =$ ..... **[4]**

(c) Calculate the 12th term of the sequence described in part (b).

..... **[2]**

16 **(a)** Find the nth term of the sequence.

5 9 13 17 21

..... **[2]**

(b) Find the nth term of the sequence.

2 6 18 54 162

..... **[2]**

17 Here are the first five terms of a sequence.

4 11 22 37 56

(a) Work out the next term of the sequence.

..... **[1]**

(b) Find an expression for the nth term of the sequence.

..... **[3]**

18 The first five terms of three sequences, A, B and C, are shown below.

Sequence A:	5	9	13	17	21
Sequence B:	2	6	20	50	102
Sequence C:	$\frac{5}{2}$	$\frac{3}{2}$	$\frac{13}{20}$	$\frac{17}{50}$	$\frac{7}{34}$

(a) Find the nth term of sequence A.

..... **[2]**

(b) **(i)** Work out the 6th term of sequence B.

..... **[1]**

(ii) Find an expression for the nth term of sequence B.

..... **[4]**

(c) **(i)** Work out the 6th term of sequence C, giving your answer as a fraction in its simplest form.

..... **[1]**

(ii) Find an expression for the nth term of sequence C.

..... **[2]**